

BriefRoom *An AI powered Assessment Centre rehearsal*

A browser room where a candidate sits with AI peers, gets interrupted in character, presents as a team, and leaves with scores plus a coach.

PROBLEM STATEMENT

Graduate hiring still uses Assessment Centres: a timed case, a group discussion with strangers, then a team presentation. People usually fail the live table, not the written case. Introverts freeze or get talked over because they almost never practise that exact setting. Mock interviews are one to one. Coaches are expensive. University services cannot sit at a live table several times a week. **The opportunity is a rehearsal room that feels like the real table:** mixed personalities, live voice, rational interruption, then a specific debrief. Candidates can fail in practice, not on assessment day.

SOLUTION OVERVIEW

BriefRoom is a browser Assessment Centre. The user picks an industry, sets three AI teammates and their personality, waits for a generated case pack, reads it in prep, then speaks on a microphone. In discussion, teammates can interrupt if it fits their character. In the last 30 seconds someone wraps the main points. Presentation then runs one by one, one point at a time, round robin, with no interruption. At the end the user gets competency scores and can highlight any line for a better reply.

A FastAPI WebSocket room owns the clock, the floor, and speaking order. LangGraph runs named jobs (case writer, interrupt jury, panelist speech, assessor, coach). Discussion and presentation use two different rules so the table does not collapse into overlapping speech. Free tier voices have no true accent yet. Accent is only written into the prompt.

USE OF AI

Coding tools. Grok designed the architecture and most of the code (LangGraph graph, FastAPI room, voice wiring, frontend). Live bugs such as wrapping too early, models blocked in this region, and TTS failures were found in runs and patched in the same loop.

Models, APIs, platforms. OpenRouter chat completions: a fast model (DeepSeek Flash) for live panelists and interrupt votes; a stronger model for the case pack and hidden assessor rubric. OpenRouter TTS for voices (Kokoro default) with browser speechSynthesis as fallback. Chrome Web Speech API for live partial transcripts so interruption can happen mid sentence. Keys stay on the server.

System prompts (examples). Panelist: “You are a real candidate. Never break character. If you interrupt, your first sentence must react to the words the person is still saying.” Interrupt jury: “Only interrupt if you strongly disagree and you can name the exact claim they just made. Default is no.” Case writer must invent a specific company and an uncomfortable trade off, and must not put a model answer in the candidate pack. Assessor scores only what is in the transcript and quotes the candidate.

IMPACT AND VALUE

Users get a new pack every run, a full transcript of a messy table, scores with quotes, and a better line for any highlighted moment. Introverts and career switchers can practise the conversation that usually fails them. Universities get a repeatable group exercise room. Firms get candidates who have already failed in rehearsal. Good use looks like this: the candidate leaves with one structural habit to change, for example open with the ask, not a life story.

REFLECTIONS

What worked. Keep LangGraph nodes as workers and the WebSocket room as conductor. Split discussion (bargain in) from presentation (ordered rotation). Inject the official remaining time into prompts instead of letting the model guess the clock. Concurrent interrupt votes plus a personality prior beat one shared “does anyone want to talk?” call.

Challenges. Interruption logic was the hard part. Early versions wrapped too early, dumped one long speech, or talked over themselves. API limits also hurt: model slugs blocked in this region, TTS errors, and free tier voices with no real accent.

Next, given more time. Scale from three demo seats to six people around one table. Add real accents with ElevenLabs tokens. Build a smoother real time multi agent pipeline. Later, try VR so the room feels closer to sitting at a real desk. Main lesson: models write the sentences; the room must own time, floor, and social rules.